BrainVLN v2

B5 · conv frontend · pooled layout · 4096 LIF/ALIF

RGB
[B, 3, 224, 224]

RGB ConvEncoder
7×7/4, 3×3/4, 3×3/2
→ 64-d

Depth
[B, 1, 224, 224]
× depth_valid

Depth ConvEncoder
same tower
masked depth

RGBDFusion
concat
Linear 128 → 64
cross-attention off

fused

history

Spatial 1536
spatial_proj(fused ‖ history)

Previous action
id 0–4
start id = 4

Action embedding
Embedding(5, 64)

EventMemory
surprise > 0.5
write = fused × gate

Episodic 1280
episodic_proj 64 → 1280

Progress
Linear(Episodic)
1280 → 1

Instruction
tokens [B, L]
token_mask

Language
masked mean → 64
subinstruction = 0

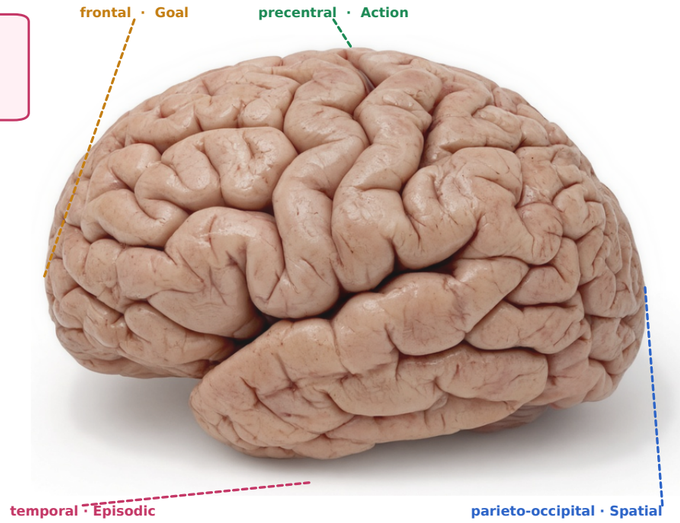
goal_proj
Linear 128 → 768

Goal 768
instruction current

Action 512
external current = 0

Readout
Linear(Action ‖ Goal)
→ 4 logits

Actions
STOP · MOVE_FORWARD
TURN_LEFT · TURN_RIGHT



SynapticGraph

Spatial	Spatial → Goal Spatial → Episodic Spatial → Action Spatial ↺ local	
Goal	Goal → Episodic Goal → Action no self-edges	
Episodic	Episodic → Goal Episodic → Action Episodic ↺ recurrent	
Action	Action ↺ recurrent no cross-population outgoing edges	

2 LIF/ALIF steps per navigation step
delay 1–4 · out-degree 48 · 80% excitatory
closed loop: action → next RGB, depth, previous action